

Redlining's Environmental Impact in Los Angeles

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Load in Required Packages

```
# Check if required packages are installed
required_packages <- c("sf", "tmap", "tidyverse", "here", "knitr", "patchwork", "paletteer")

for(pkg in required_packages){
  if (!require(pkg, character.only = TRUE)){
    stop(paste0("Package '", pkg, "' is not installed"))
  }
}

# Load required packages
suppressPackageStartupMessages({
  library(sf)
  library(tmap)
  library(tidyverse)
  library(here)
  library(knitr)
  library(patchwork)
  library(paletteer)
})
```

Reading in Data

```

# EJScreen Data
ejscreen <- sf::st_read(here::here("data","ejscreen","EJSCREEN_2023_BG_StatePct_with_AS_CNMI.

# Redlining Data
redlining <- sf::st_read(here::here("data","mapping-inequality","mapping-inequality-los-ange

# Bird observations in LA County
birds <- sf::st_read(here::here("data", "gbif-birds-LA", "gbif-birds-LA.shp"))

# Check if the geometry of objects match & change if false
if (st_crs(birds) != st_crs(ejscreen)) {
  birds <- st_transform(birds, crs = st_crs(ejscreen))
}
if (st_crs(redlining) != st_crs(ejscreen)) {
  redlining <- st_transform(redlining, crs = st_crs(ejscreen))
}

# Data for map visualization
ejscreen_subset <- ejscreen %>%
  filter(ST_ABBREV == c('CA'),
         CNTY_NAME %in% c("Los Angeles County",
                          "Orange County"))

# LA County data from ejscreen
los_angeles <- ejscreen_subset %>% filter(CNTY_NAME == "Los Angeles County")

```

Map of Redlining in Los Angeles Census Blocks

```

# Map area
custom_bbox <- c(xmin = -118.65,
                 ymin = 33.7,
                 xmax = -117.7,
                 ymax = 34.31)

# Color Palette for grades
custom_palette <- c("#939E63", "#8A9D9B", "#D3B869", "#BF7A80")

# Map of Redlining in Los Angeles County
tm_shape(redlining, bbox = custom_bbox) +

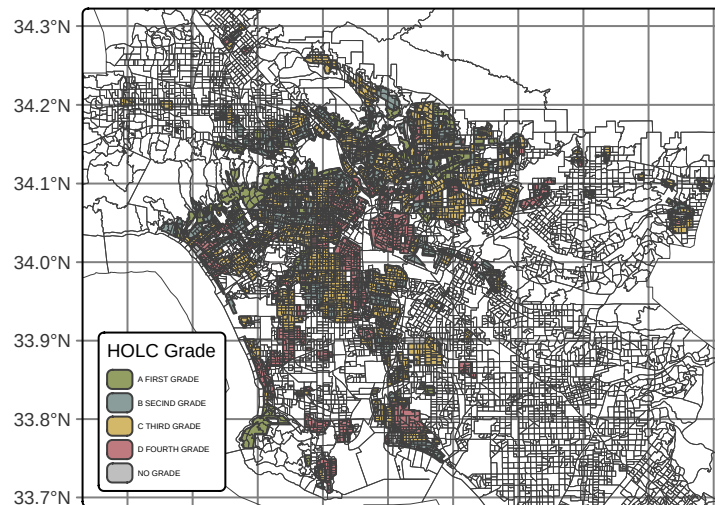
```

```

tm_graticules(crs = 4326, labels.show = TRUE) +
tm_polygons("grade",
  fill.legend = tm_legend(title = "HOLC Grade", title.size = .60, text.size = .5),
  fill.scale = tm_scale(
    values = custom_palette,
    labels = c("A FIRST GRADE", "B SECIND GRADE", "C THIRD GRADE", "D FOURTH GRADE", "NO GRADE"),
    label.na = "NO GRADE", # Change default na label
    orientation = "portrait")) +
tm_layout(legend.position = tm_pos_in(pos.h = 0.025, pos.v = 0.35),
  legend.width = 6,
  legend.height = 7.5) +
tm_title(text = "Redlining in Los Angeles Census Blocks") +
tm_shape(ejscreen_subset) +
tm_borders(lwd = .3) +
tm_options(component.autoscale = FALSE)

```

Redlining in Los Angeles Census Blocks



Summary Table of Redlining Data

```

# Join LA and redlining data
joined_data <- st_join(redlining, los_angeles) %>%
  st_drop_geometry() # Drop geometry

# Give NA values in grade column a label
joined_data$grade[is.na(joined_data$grade)] <- "None"

#.....KABLE.....
# Table showing the percent of blocks in each grade
summary_data <- joined_data %>%
  group_by(grade) %>%
  summarise(BlockCount = n()) %>% # Total count of blocks in LA
  mutate(percent = BlockCount/sum(BlockCount) * 100) %>% # Count at each site / Total
  kable(col.names = c("HOLC Grade", "Count", "Percent in Grade"),
        align = 'c', # Center text
        caption = "Percent of Blocks in Each HLOC Grade", # Title
        digits=3) # Limit to three decimal points

# Output
summary_data

```

Table 1: Percent of Blocks in Each HLOC Grade

HOLC Grade	Count	Percent in Grade
A	449	7.029
B	1239	19.396
C	3058	47.871
D	1346	21.071
None	296	4.634

Visualizations of Current Observations

```

# Create data frame with mean values of:
#   % Low Income
#   Percentile for PM2.5
#   Percentile for Low Life Expectancy
# based on the HOLC Grade
summary_data <- joined_data %>%

```

```

group_by(grade) %>% # Group by grade
# Summarize for mean values
summarize(low_income = mean(LOWINCPCT, na.rm = TRUE)*100, # Convert to whole number
           particulate_matter = mean(P_D2_PM25, na.rm = TRUE),
           life_expect = mean(P_LIFEEXPCT, na.rm = TRUE))

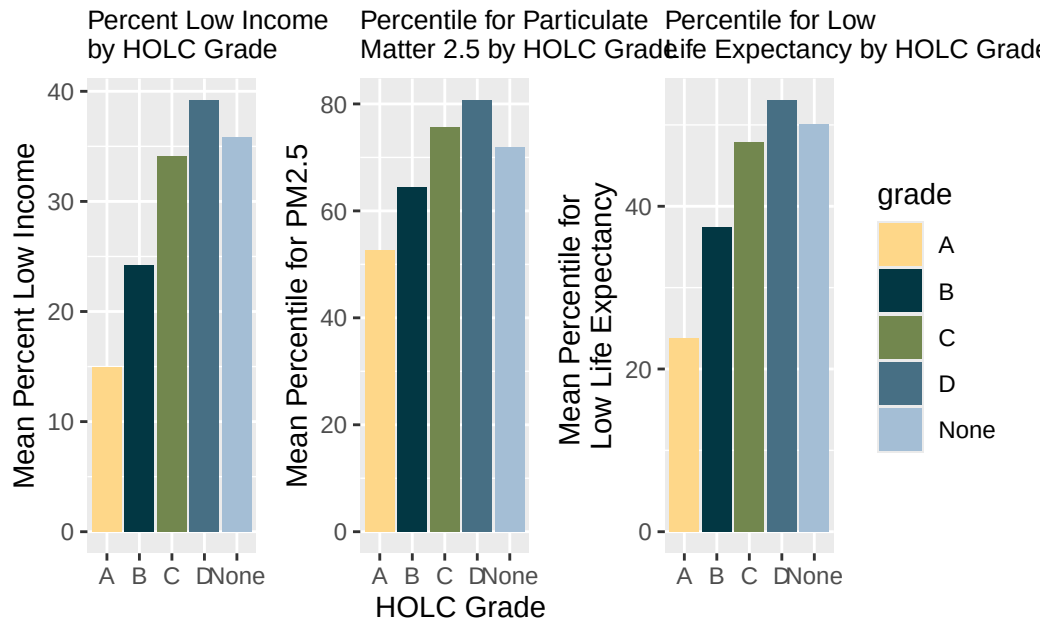
#.....Patchwork Plots.....
# Plot of Percent Low Income
p1 <- ggplot(summary_data, aes(x = grade, y = low_income, fill = grade)) +
  geom_col() +
  labs(title = "Percent Low Income \nby HOLC Grade",
       y = "Mean Percent Low Income",
       x = "HOLC Grade") +
  scale_fill_paletteer_d("nationalparkcolors::Acadia") # Color Palette

# Plot of PM2.5 Percentile
p2 <- ggplot(summary_data, aes(x = grade, y = particulate_matter, fill = grade)) +
  geom_col() +
  labs(title = "Percentile for Particulate \nMatter 2.5 by HOLC Grade",
       y = "Mean Percentile for PM2.5",
       x = "HOLC Grade") +
  scale_fill_paletteer_d("nationalparkcolors::Acadia") # Color Palette

# Plot of Low Life Expectancy Percentile
p3 <- ggplot(summary_data, aes(x = grade, y = life_expect, fill = grade)) +
  geom_col() +
  labs(title = "Percentile for Low \nLife Expectancy by HOLC Grade",
       y = "Mean Percentile for \nLow Life Expectancy",
       x = "HOLC Grade") +
  scale_fill_paletteer_d("nationalparkcolors::Acadia") # Color Palette

# Merge Plots Horizontally
p1 + p2 + p3 + plot_layout(guides = "collect", axes = "collect_x") &
  theme(plot.title = element_text(size = 10)) # Resize plot titles

```



Discussion

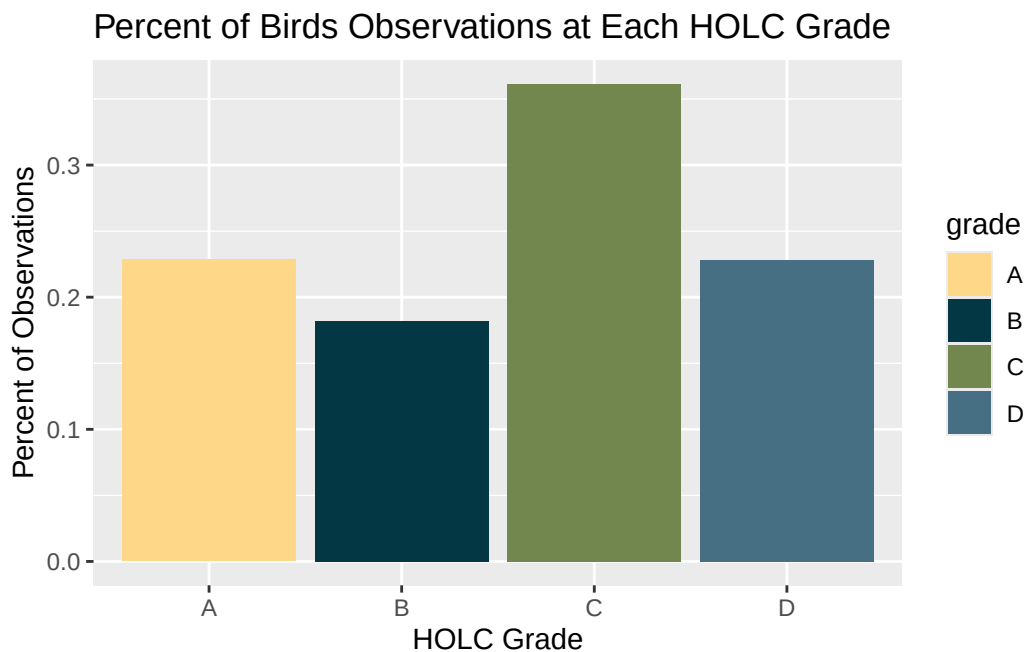
The bar charts above indicate that neighborhoods which were assigned lower HOLC grades have worse environmental and socioeconomic conditions than areas with higher grades. In all three graphs there is a consistent increase in the parameter of interest as the grade changes from A to D indicating that the stratification of the population caused by this grading can still be observed in the modern day. Additionally, neighborhoods which had no assigned grade tended to fall between the C and D grades on the bar graphs which indicates that they could have been indirectly affected by the redlining policies of the time. Overall, it seems that the inequality caused through redlining has been maintained despite its banning seventy years ago, showing the devastating long term impact these policies had.

Legacy of Redlining in Biodiversity Observations

```
if (st_crs(birds) != st_crs(redlining)) {
  warning("CRS of objects do not match")
}
```

```
# Data frame with percent of bird observations at each site
birds_joined <- st_join(redlining, birds, join = st_contains) %>%
  st_drop_geometry() %>% # Drop geometry since it's no longer needed
  filter(!is.na(grade)) %>% # Remove grades with NA values
  group_by(grade) %>%
  summarize(percent_observed = n()) %>% # Get number of observations
  mutate(percent = percent_observed/sum(percent_observed)) # Make new columns with percent of

# Bar plot of percent of bird observations at each HOLC site
ggplot(birds_joined, aes(x = grade, y = percent, fill = grade)) +
  geom_col() +
  # Label axis
  labs(title = "Percent of Birds Observations at Each HOLC Grade",
       y = "Percent of Observations",
       x = "HOLC Grade") +
  scale_fill_paletteer_d("nationalparkcolors::Acadia") # Color palette
```



Discussion

Our findings differed from the paper’s expectations: we found that areas in the C and D grade areas having a higher contribution to the total number of observations than A and B graded areas. This is likely due to the paper considering the density of sampling rather than the sheer number of samples, which is what we did. While C and D grade areas did have a higher number of observations, they also constitute a significantly higher percentage of Los Angeles total land area than A or B areas which can be seen in the map above.